

1. Filesystem

```
hecker@Windows: ~/Project_Os/Week_1
hecker@Windows:~$ mkdir Project_Os
hecker@Windows:~$ cd Project_Os
hecker@Windows:~/Project_Os$ mkdir Week_1
hecker@Windows:~/Project_Os$ cd Week_1
hecker@Windows:~/Project_Os/Week_1$ echo hello > file_1
hecker@Windows:~/Project_Os/Week_1$ echo hi > file_2
hecker@Windows:~/Project_Os/Week_1$ ln file_1 hard
hecker@Windows:~/Project_Os/Week_1$ ln -s file_2 soft
hecker@Windows:~/Project_Os/Week_1$ ls -li
total 12
11154790 -rw-rw-r-- 2 hecker hecker 6 Mar 30 19:47 file_1
11154792 -rw-rw-r-- 1 hecker hecker 3 Mar 30 19:47 file_2
11154790 -rw-rw-r-- 2 hecker hecker 6 Mar 30 19:47 hard
11154807 lrwxrwxrwx 1 hecker hecker 6 Mar 30 19:47 soft -> file_2
hecker@Windows:~/Project_Os/Week_1$ rm file_1
hecker@Windows:~/Project_Os/Week_1$ ls -li
total 8
11154792 -rw-rw-r-- 1 hecker hecker 3 Mar 30 19:47 file_2
11154790 -rw-rw-r-- 1 hecker hecker 6 Mar 30 19:47 hard
11154807 lrwxrwxrwx 1 hecker hecker 6 Mar 30 19:47 soft -> file_2
hecker@Windows:~/Project_Os/Week_1$ cat hard
hello
hecker@Windows:~/Project_Os/Week_1$ cat soft
hi
hecker@Windows:~/Project_Os/Week_1$

hecker@Windows:~/Project_Os/Week_1$ echo test > original
hecker@Windows:~/Project_Os/Week_1$ ln -s original soft_test
hecker@Windows:~/Project_Os/Week_1$ rm original
hecker@Windows:~/Project_Os/Week_1$ cat soft_test
cat: soft_test: No such file or directory
hecker@Windows:~/Project_Os/Week_1$
```

2. Permissions

a. File

```
hecker@Windows: ~/Project_Os/Week_1/Permissions
hecker@Windows:~/Project_Os/Week_1$ mkdir Permissions
hecker@Windows:~/Project_Os/Week_1$ cd Permissions
hecker@Windows:~/Project_Os/Week_1/Permissions$ echo data > file
hecker@Windows:~/Project_Os/Week_1/Permissions$ chmod 000 file
hecker@Windows:~/Project_Os/Week_1/Permissions$ cat file
cat: file: Permission denied
hecker@Windows:~/Project_Os/Week_1/Permissions$ chmod 644 file
hecker@Windows:~/Project_Os/Week_1/Permissions$ cat file
data
hecker@Windows:~/Project_Os/Week_1/Permissions$
```

b. Directory

```
hecker@Windows: ~/Project_Os/Week_1/Permissions
hecker@Windows:~/Project_Os/Week_1$ cd Permissions
hecker@Windows:~/Project_Os/Week_1/Permissions$ mkdir directory
hecker@Windows:~/Project_Os/Week_1/Permissions$ echo hi > directory/a
hecker@Windows:~/Project_Os/Week_1/Permissions$ chmod -x directory
hecker@Windows:~/Project_Os/Week_1/Permissions$ ls directory
ls: cannot access 'directory/a': Permission denied
a
hecker@Windows:~/Project_Os/Week_1/Permissions$ chmod +x directory
hecker@Windows:~/Project_Os/Week_1/Permissions$ ls directory
a
hecker@Windows:~/Project_Os/Week_1/Permissions$
```

3. Backup

```
hecker@Windows: ~/Project_Os/Week_1
hecker@Windows:~/Project_Os/Week_1$ mkdir data
hecker@Windows:~/Project_Os/Week_1$ echo A > data/a
hecker@Windows:~/Project_Os/Week_1$ echo B > data/b
hecker@Windows:~/Project_Os/Week_1$ du -sh data
12K    data
hecker@Windows:~/Project_Os/Week_1$ tar -cvf backup.tar data
data/
data/a
data/b
hecker@Windows:~/Project_Os/Week_1$ gzip backup.tar
hecker@Windows:~/Project_Os/Week_1$ du -sh backup.tar.gz
4.0K    backup.tar.gz
hecker@Windows:~/Project_Os/Week_1$ tar -tf backup.tar.gz
data/
data/a
data/b
hecker@Windows:~/Project_Os/Week_1$
```

4. Aliases

```
hecker@Windows: ~/Project_Os/Week_1
hecker@Windows:~/Project_Os/Week_1$ alias ll = 'ls -l'
alias ll='ls -alf'
bash: alias: =: not found
bash: alias: ls -l: not found
hecker@Windows:~/Project_Os/Week_1$ alias rm = 'rm -i'
bash: alias: rm: not found
bash: alias: =: not found
bash: alias: rm -i: not found
hecker@Windows:~/Project_Os/Week_1$ ll
total 28
drwxrwxr-x 4 hecker hecker 4096 Mar 30 19:58 ./
drwxrwxr-x 3 hecker hecker 4096 Mar 30 19:46 ../
-rw-rw-r-- 1 hecker hecker 184 Mar 30 19:58 backup.tar.gz
drwxrwxr-x 2 hecker hecker 4096 Mar 30 19:58 data/
-rw-rw-r-- 1 hecker hecker  3 Mar 30 19:47 file_2
-rw-rw-r-- 1 hecker hecker  6 Mar 30 19:47 hard
drwxrwxr-x 3 hecker hecker 4096 Mar 30 19:55 Permissions/
lrwxrwxrwx 1 hecker hecker  6 Mar 30 19:47 soft -> file_2
hecker@Windows:~/Project_Os/Week_1$ touch test
hecker@Windows:~/Project_Os/Week_1$ rm test
hecker@Windows:~/Project_Os/Week_1$ \rm test
rm: cannot remove 'test': No such file or directory
hecker@Windows:~/Project_Os/Week_1$
```